

Technical Document #705: Data Engineering - Comprehensive Overview

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Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Key Takeaways:

- Stream processing handles continuous data streams in real-time.
- Data lineage tracks the origin and movement of data through systems.
- Partitioning divides data into smaller, more manageable pieces.